GGG-S-282B
November 13, 1969
SUPERSEDING
Int. Fed. Spec. GGG-S-00282d(NAVY-Ships)
October 26, 1966 and
Fed. Spec. GGG-S-282b
January 31, 1958
(See 6.4)

FEDERAL SPECIFICATION

SHEARS, METAL CUTTING, ELECTRIC (PORTABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric metal cutting shears.

1.1.1 Federal specification coverage. Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specification, but cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Portable electric shears shall be of the following types, as specified (see 6.2):

Type I - Shears, single shear action.

Sizes 1, 2, and 3 (see table I).

Type II. Shears, double shear action.

Size 4 (see table II).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-T-152 - Treatment, Moisture-and-Fungus-Resistant of Communications, Electronics and Associated Electrical Equipment.

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electric and Associated Electrical Equipment.)

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1337 - Electric Portable Hand Tools General Suppression System Design Requirements.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issues in effect on date of invitation for bids, or request for proposal shall apply:

National Bureau of Standards (NBS) Handbook:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American National Standards Institute (ANSI), Inc. Standards:

B3.5 - Tolerances for Ball and Roller Bearings.
B3.6 - Boundary dimensions for Ball and Roller Bearings.
B3.8 - Mounting Ball and Roller Bearings, Specification for.

(Application for copies should be addressed to the American National Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

The Anti-Friction Bearing Manufacturers' Association (AFBMA) Standards:

Section No. 5 - Bearing Identification Code for Anti-Friction Ball and Roller Bearings.

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers' Association, Inc., 60 East 42nd Street, New York, N.Y. 10017.)

Underwriters' Laboratories (UL), Inc. Standard:

No. 45 - Electric Tools.

(Applications for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 333 Pfingsten Rd., Northbrook, Illinois 60062; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of portable electric shears which are otherwise in accordance with the requirements of this specification.

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3.2.1 Cutting characteristics. Shears shall be capable of making straight, irregular contour and circular cuts in ferrous and non-ferrous materials of sheet and plate form. Shears shall make clean cuts. Burrs or ragged edges are not acceptable.

3.2.2 Safety. Shears shall be mechanically and electrically safe. Safety devices shall be provided to protect the operator from accidental contact with rotating, reciprocating or such moving parts which might be hazardous. Electrical components and motor shall be completely enclosed, except for ventilation openings conforming to UL45.

3.2.3 Lubrication. Bearings, except sealed bearings which are prelubricated, gears and moving parts shall be assembled with lubricating medium. Provision for lubricant replacement shall be provided. Lubricant shall not leak from the gear casing or enter the motor compartment.

3.2.4 Interchangeability. All parts from the same manufacturer shall be manufactured to standards which will permit interchangeability of similar parts without modification or adjustment.

3.3 Material. Materials used in fabrication of shears shall be new, of uniform quality and be produced according to standard practices of reputable manufacturers producing equipment which will meet the requirements of this specification.

3.4 Construction. Shears shall be complete so that when connected to the specified sources of power, they can be used for any designed operation. Construction shall be free from any characteristics or defects that will prevent the machine from passing the examination and tests in section 4.

3.4.1 Castings. All castings shall be free from defects, scale, or mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or metallic pastes be used on casings for reclaiming any part.

3.4.2 Surfaces. Surfaces of all castings, forgings, and molded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled. Surfaces shall be finished according to the manufacturer's standard practice.

3.4.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.4.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means of preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or permanently deformed. Threaded parts shall be in the inch system and shall conform to H-28.

3.5 Type I Shears, single shear action. Type I shears shall be of the manually held and guided type, with a pair of blades, one to be a stationary blade, the other to be the cutting blade with a reciprocating movement. Type I, sizes 1, 2, and 3 shall be similar to figs. 1 or 1a, as applicable, and shall comply with the requirements of table I for the size specified (see 6.2).

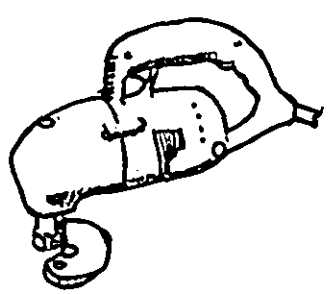


FIGURE 1. Shears, electric, portable, type I, (sizes 1, 2, and 3).

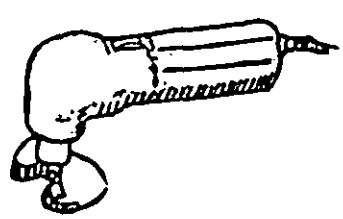


FIGURE 1a. Shear, electric, portable, type I (size 1 only).

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TABLE 1. Type I requirements

Size	Capacity		Cutting rate (Min.)	Minimum cutting radius at least	Overall length (Max.)	Weight without cable (Max.)
	Manufacturers gage	Steel thickness (approx.)				
		Inch	Ft/Min.	Inch	Inches	Pounds
1	18 -	0.0478	15	1	12	10
2	16 -	0.0598	15	1-1/2	14	12
3	12 -	0.1046	15	2-1/2	15	16

3.5.1 Components, Type I Shear. Type I shears furnished under this specification shall consist of a motor, switch, cable and plug, wiring, housings, gears and shafts, bearings, cutting blades, yoke, and handle.

3.5.1.1 Motor. The motor shall be a continuous duty universal series wound or a compensated series wound designed to operate on either 115 volt, direct current or 115 volt, single phase, 60 cycle alternating current. Motors shall comply with the requirements of UL45 for electric tools.

3.5.1.2 Switch. Shears shall be provided with a conveniently located switch of quick break design. Self-releasing switches shall return to the "OFF" position when pressure is removed and shall be provided with a locking pin for locking in "ON" position. Switches other than self-releasing type shall be marked to indicate "ON" and "OFF" positions. Switches shall comply with the requirements of UL45.

3.5.1.3 Wiring. All wiring, flexible cord, attachment plug cap and grounding means shall be in accordance with UL45. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), a minimum free length of 9 feet measured from point of entry into the shears.

3.5.1.4 Housings. Housings shall be of a material of suitable characteristics to withstand stresses imposed by normal usage. Housings shall include the motor case, gear and cam and reciprocating mechanism housing.

3.5.1.5 Gears and shafts. Gears shall be alloy steel. Where worm gearing is used, the gearing shall consist of a steel worm and bronze worm wheel. Gears shall have machine cut or formed teeth and be designed and supported to efficiently transmit imposed loads. Shafts shall be of alloy steel supported by at least two bearings to prevent deflection. Gears and shafts shall be fully enclosed.

3.5.1.6 Bearings. Armature shafts shall be mounted in ball or roller bearings. Rotating shafts, other than the armature shaft, shall be mounted in ball, roller or needle roller bearings. Reciprocating shafts may be mounted in a bronze sleeve bearing. Roller or needle roller bearings which are designed primarily for radial loads shall not be used to absorb thrust loads. Steel inserts may be omitted in the mounting of bearings when sufficient contact area and a suitable fit between bearing outer diameter and housing is provided. Bearings shall be of the dustproof sealed type or shall be mounted in a dustproof housing. Ball, roller, and needle roller bearings shall conform to ANSI B3.5 for tolerances, B3.6 for dimensions and B3.8 for mounting requirements for the type and size bearings used and (AFBMA Standard Section No. 5) for identification.

3.5.1.7 Cutting blades. Cutting blades shall be of alloy tool steel, heat-treated to a hardness of 57-60 on the Rockwell C scale. The reciprocating cutter blade shall produce a single shear action and shall be clearly visible to the tool operator to enable the operator to follow the line to be cut. Cutter blades shall be mounted for convenient adjustment or replacement.

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3.5.1.8 Yoke. The yoke shall contain the stationary cutter blade and shall be attached to the reciprocating cutter housing. The yoke shall be of a material and cross-sectional area suitable to maintain cutter blade adjustment and alignment.

3.5.1.9 Handles. Type I shears shall be equipped with a comfortable closed double grip handle similar to either fig. 1, (sizes 1, 2, or 3) or a body type handle similar to fig. 1a, (size 1).

3.6 Type II Shears, double shear action. Type II, size 4 shears shall be of the hand guided type, with one cutting blade pivoted to operate between two stationary blades producing a double shear action. Type II shears shall be similar to fig. 2 and shall comply with the requirements of table II.

TABLE II. Type II requirements

Size	Capacity		Cutting rate (Min.)	Minimum cutting radius at least	Overall length (Max.)	Weight (Max.)
	Manufacturers gage	Steel (thickness) (approx.)				
		Inch	Ft./Min.	Inches	Inches	Pounds
4	18	0.0478	15	3	13	5

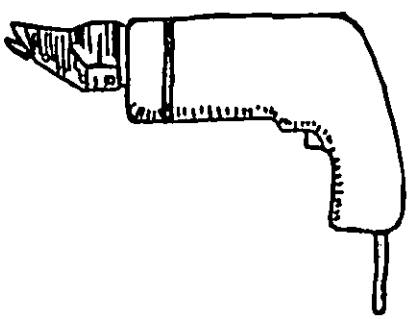


FIGURE 2. Shear, electric, portable, type II.

3.6.1 Components, Type II Shears. Components for type II shears shall comply with the applicable requirements for Type I, except for swivel head, cutting blades, and handle. Type II shears shall consist of motor switch, cable and plug, wiring, housings, gears and shafts, bearings, swivel head, cutting blades, and handle.

3.6.1.1 Swivel head. Type II shears shall be provided with a swivel head having a locking device to fix the head in one of several positions. The swivel head shall contain the cutting blades.

3.6.1.2 Cutting blades. Cutting blades shall be mounted for convenience of adjustment and repair. Blades shall be alloy tool steel, heat treated to a hardness of 55-60 on the Rockwell C scale. The reciprocating cutter blade shall be visible to the tool operator during operation of the tool to enable the operator to follow the line to be cut.

3.6.1.3 Handle. Type II shears shall have a pistol grip handle similar to fig. 2.

3.7 Electromagnetic interference. When specified (see 6.2), each shear shall be electromagnetic interference free in accordance with the requirements of MIL-STD-1337 for class II B equipment.

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3.8 Moisture-and-fungus-resistant-treatment. When specified (see 6.2), electrical components of shears covered by this specification shall be moisture-and-fungus-resistant treated with material conforming to and applied as required by MIL-T-152 and MIL-V-173 as applicable.

3.9 Identification plate. A corrosion resistant nameplate, permanently and legibly marked to include the following information shall be securely attached to the shear:

a. Manufacturer's name, model or serial number.
b. Mild steel cutting capacity.
c. Voltage.
d. Rated-load amperes.

3.10 Instruction book. Unless otherwise specified (see 6.2), the manufacturer shall furnish one copy of his standard instruction book and parts list with each shear.

3.11 Accessories. Each shear shall be furnished with one extra set of cutting blades, and such accessories and other extra parts as normally furnished commercially.

3.12 Workmanship. Workmanship of the shears shall be of the quality prevailing among manufacturers normally producing shears of the type and sizes specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Classification of inspection. Inspection shall be classified as quality conformance inspection.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All shears of the same type and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level I.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample shear selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

a. Examination (4.4.1).
b. Tests (4.5.1 thru 4.5.5).
c. Inspection of preservation, packaging, packing, and marking for shipment and storage (4.6).

4.4.1 Examination. Each shear selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table III. Acceptable quality level (AQL) to be as indicated. Nonconforming shears shall not be offered for delivery and if the non-conforming shears in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

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Table III. Classification of defects

Categories	Defects
Critical:	AQL 0.10 Percent defective
001	Evidence of electric shock or fire hazard
002	Moving parts improperly guarded with guards which do not interfere with normal operation of shears.
Major:	AQL 2.5 Percent defective
101	Shears not of size specified
102	Component parts missing, damaged, defective or not as specified.
103	Parts not interchangeable.
104	Motor and electrical components not as specified.
Minor:	AQL 4 Percent defective
201	Dimensions and weight not as specified.
202	Identification, nameplate illegible or not permanent.
203	Improper workmanship, not smooth finished.

4.4.2 Check tests. When required by the cognizant technical agency, sample shears of the quantity, type, and size specified shall be subjected to check tests to determine minimum requirements of the governing specifications. Sample shears shall be tested using procedures similar to those specified in 4.5.2 through 4.5.7.

4.4.3 Tests for quality conformance. Each sample selected in accordance with 4.3.2 shall be tested for compliance with the requirements of this specification. AQL shall be 2.5 percent defective. Nonconforming shears shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the samples tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.5.2 thru 4.5.5.

4.5 Tests.

4.5.1 Voltages, current, frequency, temperature and rated local conditions shall conform to UL45 requirements. Voltages are r.m.s. values unless otherwise stated.

4.5.2 Electromagnetic interference test report. If electromagnetic interference suppression is required (see 6.2), the manufacturer shall provide certification that the electromagnetic suppression network in the shears offered is identical to that of the shears previously tested and approved.

4.5.2.1 The electromagnetic interference suppression system shall be disconnected during the tests of 4.5.3.

4.5.3 Motor tests. Motors shall be tested in accordance with UL45 to determine compliance with the requirements specified for:

a. Continuity of grounding connections.
b. Insulation resistance.
c. Dielectric strength.

4.5.4 Hardness test, cutting blades. Hardness of cutting blades shall be determined. Hardness for type I shear cutting blades shall be not less than 57 nor more than 60 on the Rockwell C scale. Type II shear cutting blade hardness shall be not less than 55 nor more than 60 on the Rockwell C scale.

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4.5.5 Operational tests. Sample shears selected shall be operated 20 minutes at no load operation. After the no load operational period rated load conditions (rated current, torqueing and voltage shall apply in conformance with UL45 requirements). The shears shall be operated at rated-load for a period of 20 minutes. During this test the shears shall be checked for switch operation and loss of lubricant. Leakage current shall be determined and shall not exceed (1) milliampere.

4.5.6 Performance test. Type I, size 1 and Type II, size 4 shears shall cut 15-feet of 18-gage commercial quality cold rolled sheet steel in a period of one minute. Type 1, size 2 shears shall cut 15-feet of 16-gage cold rolled steel in a period of one minute. Type 1, size 3 shears shall cut 15-feet of 12-gage cold rolled steel in a period of one minute. Cutting radius shall conform with requirements of tables I and II for the size shear specified. Clean cuts, burrs, ragged edges or distortion of work shall be observed. Nonconforming shears shall be rejected.

4.5.7 Motor current and temperature test. Motors shall be tested in accordance with UL45 to determine compliance with the requirements specified for current input and temperature rise.

4.6 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.

5.1 Preservation, packaging, packing, and marking shall be in accordance with the requirements of MIL-H-15424 for portable electric grinders. Levels of preservation, packaging, and packing shall be as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. In addition to any special marking required by the contract, or order, or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. In addition to any special marking specified in the contract or order (see 6.2), each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Shears as furnished under this specification are hand guided portable electric tools used for straight or contour cutting of ferrous and non-ferrous sheets by a scissors-like shearing action.

6.2 Ordering data. Purchasers should select the perferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1, 3.5, and 3.6).
(c) Length of cord, if different (see 3.5.1.3).
(d) Electromagnetic interference, when specified (see 3.7 and 4.5.2).
(e) Moisture-and-fungus-resistant treatment when required (see 3.8).
(f) Instruction book requirements, if different (see 3.10).
(g) Levels of preservation, packaging, and packing required (see 5.1).

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MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - EL, GL
Navy - SH
Air Force - 84, 85
DSAH - GS

User activities:

Army - ME
Navy - CG, YD

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST

GSA - FSS

Project No. 5130-0063

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